Indoor Localization in Smart Environments

Ambient Intelligence and Domotics

Master Degree: Artificial Intelligence for Science and Technology

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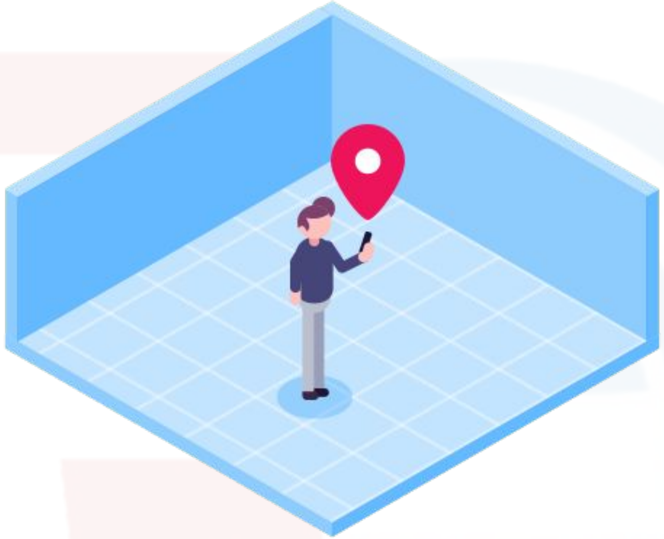
Indoor Positioning Systems

Indoor Positioning Systems (IPS)

- ***Indoor Localization*** is the process of obtaining the ***location*** of a user or a device in an indoor environment.
  - Possible thanks to the proliferation of mobile/wearable devices with ***wireless communication capabilities***
- Systems based on Indoor Localization are also called ***Indoor Positioning Systems (IPS).***
- Indoor Location is an important ***context information*** for Ambient Intelligence systems



Source: https://rannlab.com/indoor-positioning/

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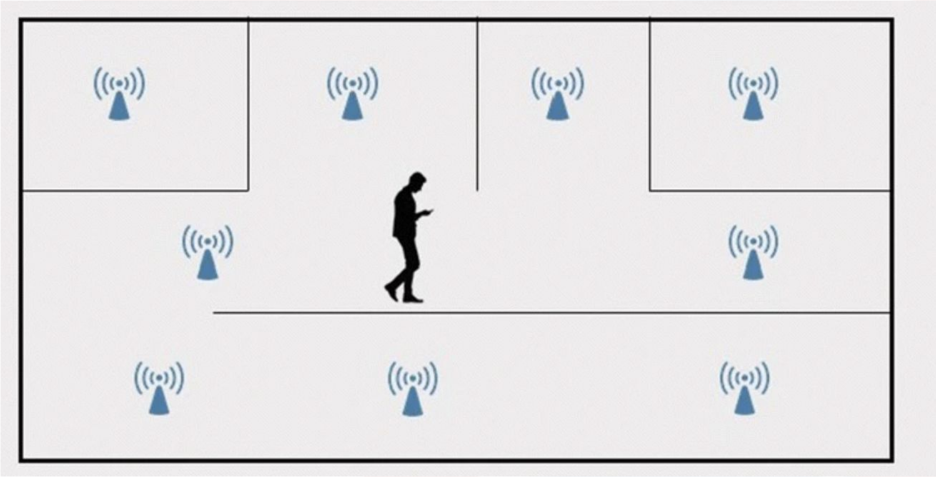
The problem

- Given:
  - a map of the building (including walking areas and obstacles)
  - a reference coordinate systems
- The goal of IPS is to determine the coordinates (x,y) in the map where an entity (e.g., a person, an object, a robot) is positioned at a specific time instant
  - it is usually framed as a ***regression*** problem



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IPS for Indoor Navigation

- Indoor positioning systems may be used to provide navigation instructions in smart buildings:
  - the indoor location of the user is continuously detected
  - the shortest indoor path to destination is computed

Source: https://www.boniglobal.com/en/project/indoor-navigation/

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IPS for Smart-Homes

- ***Ambient Assisted Living (AAL)*** applications require to monitor the behavior of elderly subjects in their homes
- Understanding the location of the subject and his/her indoor trajectories is crucial
- Particularly challenging in multi-inhabitant settings

Source: https://www.iotworlds.com/the-best-smart-home-device-to-assist-the-elderly-and-disabled/

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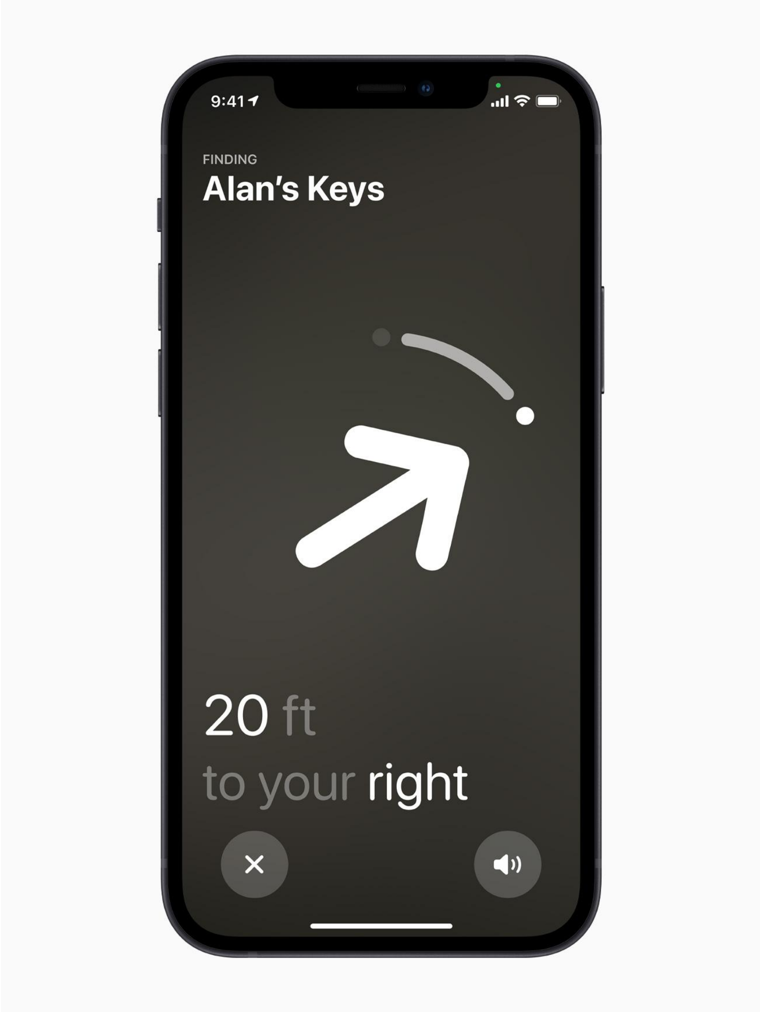
IPS for Object Localization

- As recently proposed by commercial solutions, IPS technologies may be also used to locate objects



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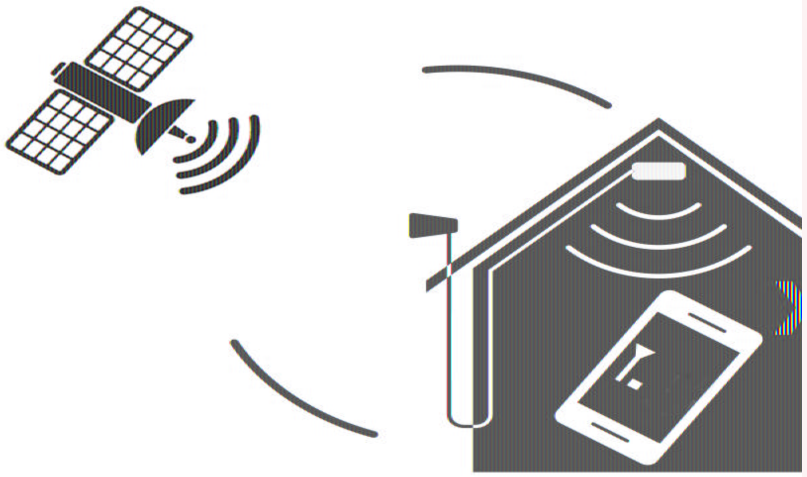
Why not simply using GPS?

- While GPS is widely used for ***outdoor localization***, it is not suitable indoors!
- The **strength** of GPS signal is low, and indoor structures (e.g., roof, walls, objects) attenuates and scatters it.
- Average error: from 5 to 10 meters!



Source: https://www.redpointpositioning.com/blog-gps/

Indoor VS Outdoor: Differences

- Coverage area is reduced (e.g., a building)
- Less impact of weather phenomena
- Availability of resources in the environment
  - e.g., electricity, Internet, …
- Slower movement speed
  - typically walking speed
- Buildings are usually ***private***
  - planimetry is not necessarily public…

Active VS Passive Indoor Positioning

- **Active methods**: the device computes the position while moving, possibly relying on sensors in the environment.
  - the ones we will see in this class!

- **Passive methods**: a distributed system computes the position, without the (explicit) cooperation with the monitored subject.
  - e.g., using cameras

Characteristics of a IPS system

Dimensions of a IPS system

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Accuracy

- Degree of conformance of an estimated position at a given time, to the value, expressed for the vertical and horizontal components
- Most common approaches
  - Root Mean Squared Deviation (RMSD) - if data distribution is normal

$$\sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(\widehat{\mathbf{P}}_i - \mathbf{P}_i\right)^2}$$

  - Average Absolute Position Deviation - more robust to outliers

$$\frac{1}{n}\sum_{i=1}^{n}\left|\widehat{\mathbf{P}}_i - \mathbf{P}_i\right|$$

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Coverage

- Spatial extension where the system performance must be guaranteed
  - **Local Coverage:** small, well-defined and not extendable area
  - **Scalable Coverage:** ability of increasing the covered area with additional hw
  - **Global Coverage:** worldwide (only applicable to GPS)

Update Rate

- The frequency at which positions are computed
  - **Periodic:** regular update, specified at some unit (e.g., Hz)
  - **On request:** triggered by user or by remote device
  - **On event:** updated when a specific event occurs (e.g., temperature sensor exceeds a threshold)

System Latency

- The delay at which information are available

  - **Real-time:** the delay should not perceived by the user.
    - e.g., crucial for indoor navigation

  - **Best effort:** the system aims at providing the answer as soon as possible based on underlying technologies

  - **Post processing:** time of delivery not required, the analysis is done after data collection

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Data Output

- Based on the application, the output may differ
  - Position
  - Heading/Bearing
  - Acceleration
  - Speed/velocity

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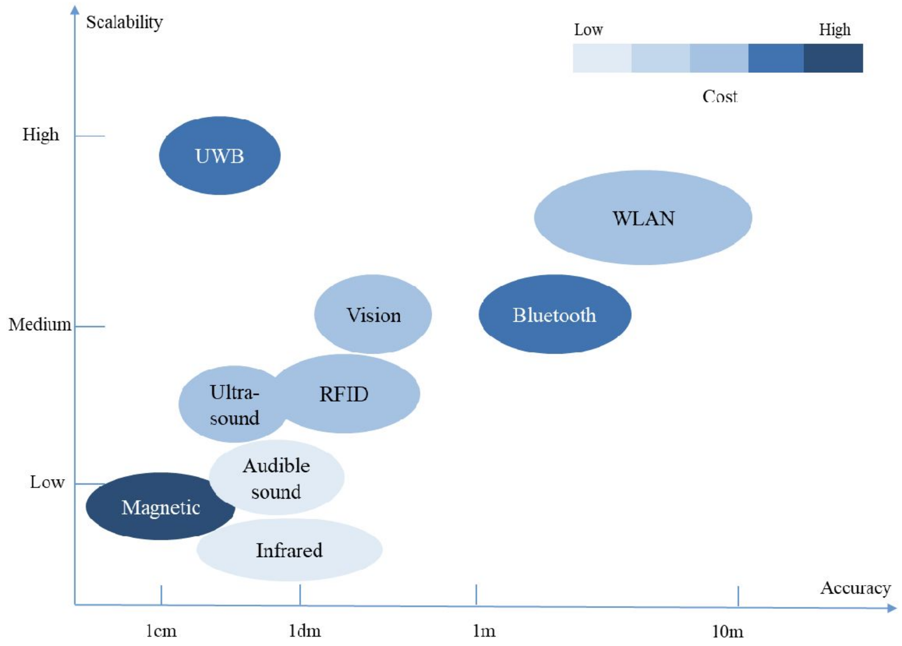
IPS Technologies

- For IPS, there is no overall solution based on a single technology
  - high diversity of solutions!

- IPS systems require dedicated local infrastructures

- Several parameters to take into account



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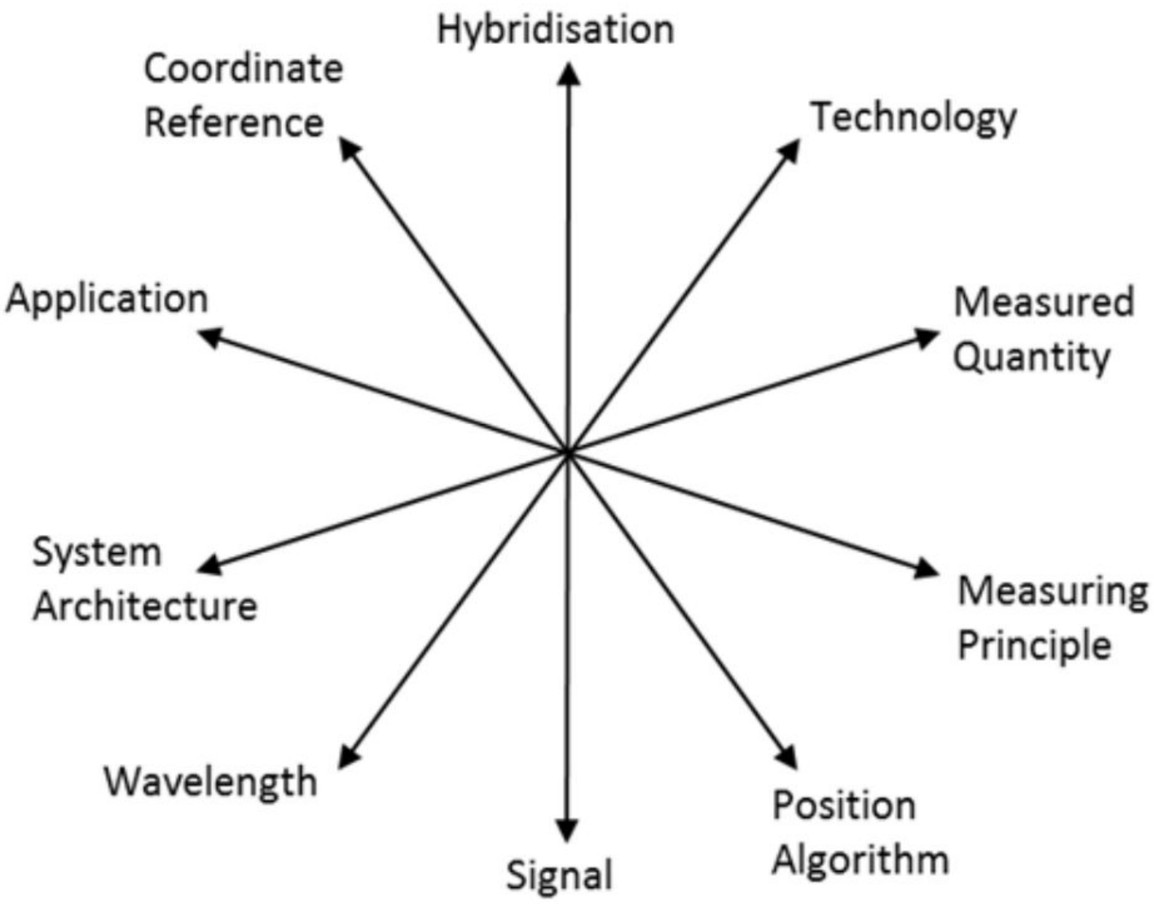
Parameters of IPS technologies



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Radio Technologies for IPS

Radio Communication Technologies

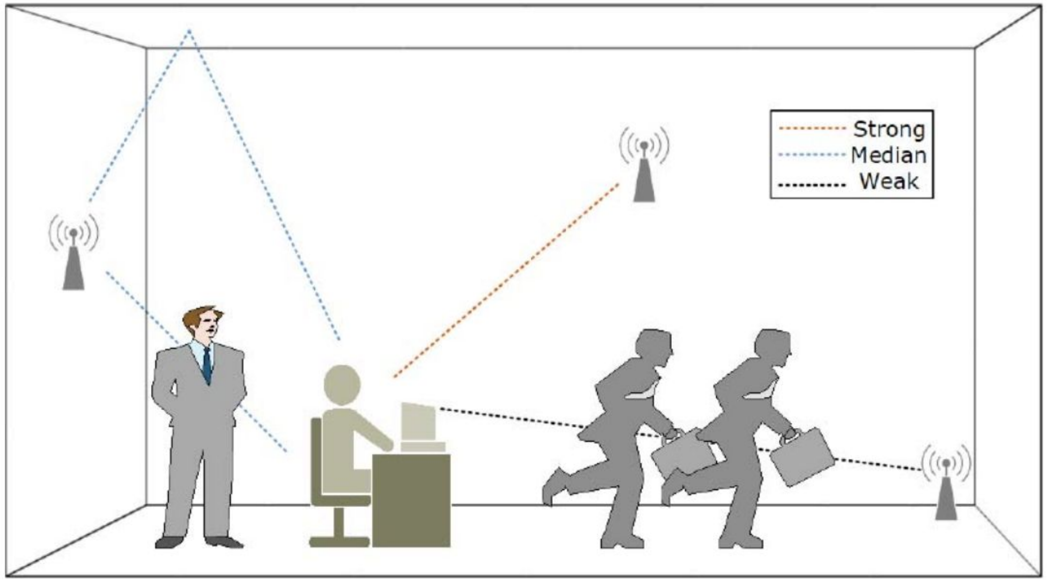
- The majority of the proposed solutions are based on radio communication technologies (e.g., WiFi, Bluetooth)

- Such technologies are used to compute the position of a *user* or a device (e.g., smartphone, smartwatch)

- One or more *antennas* in the indoor environment are required

Major Challenges

- Signal reflection on walls/furniture
- Density of obstacles
- Dynamic environments
  - e.g., presence of people, opening/closing doors, …
- Precision requirement
  - ideally, the positioning error should be below 1 meter for many applications!



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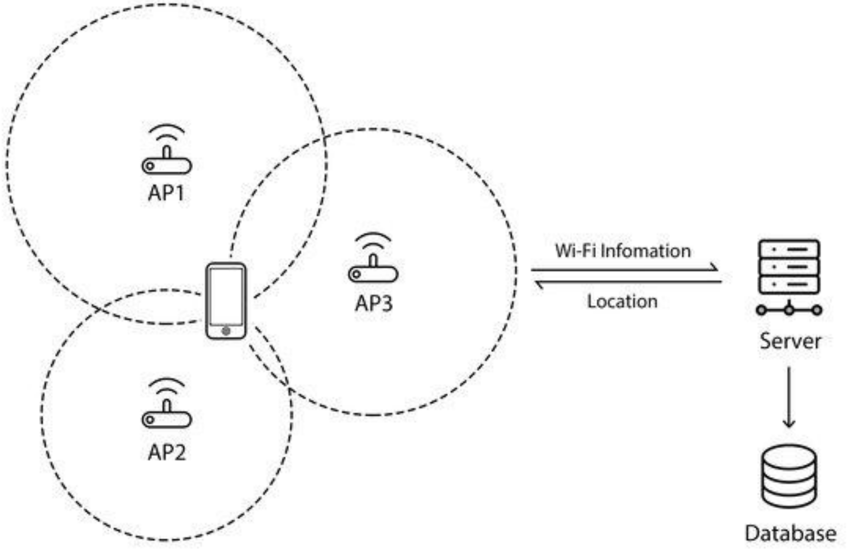
WiFi (802.11 Standard)

- Widely adopted technology
  - many devices are WiFi enabled
- Not requirements for ad-hoc infrastructures
  - many WiFi APs are already everywhere
- However, WiFi APs are normally deployed for communication rather than localization purposes
  - ad-hoc algorithms are needed to improve their localization accuracy



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Bluetooth Low Energy (BLE)

- More accurate and requiring less energy consumption w.r.t. WiFi
- Use of standard protocols (iBeacons, Eddystone)

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BLE Beacons

- Small transmitters continuously broadcasting their identifier to nearby devices
  - universally unique identifier (UUID)
- Initially conceived for proximity, they are extensively adopted for localization
- More accurate than WiFi, but deployment and maintenance is more challenging

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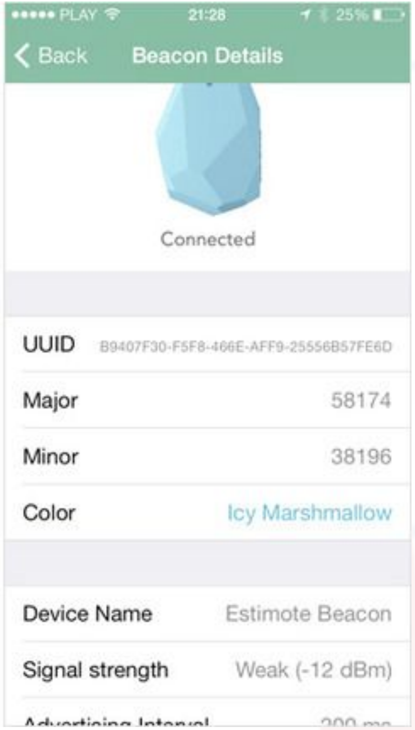
BLE beacons (cont.)



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Deploying BLE beacons

- The deployment of BLE beacons is challenging
- Position:
  - how many beacons?
  - where?
  - at which height?
- Battery:
  - beacons are equipped with a small battery requiring periodic replacements

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Ultra WideBand (UWB)

- Short pulses transmitted over a large bandwidth
- Low duty cycle
  - reduced power consumption
- Several advantages w.r.t. WiFi and BLE:
  - Less affected by interference of other signals
  - The signal can penetrate obstacles (e.g., walls)
  - Less sensitive to multipath effect
- Drawbacks:
  - requires costly and complex infrastructure
  - still not a standard deployed in personal devices

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UWB anchors and tags

- Similarly to BLE beacons, several UWB anchors should be installed in the environment in fixed and known positions
- The UWB tag is a ***mobile device*** communicating with anchors
- Anchors and tag communicate to infer the position of the tag
  - it is more accurate than BLE but more costly

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UWB enabled mobile devices

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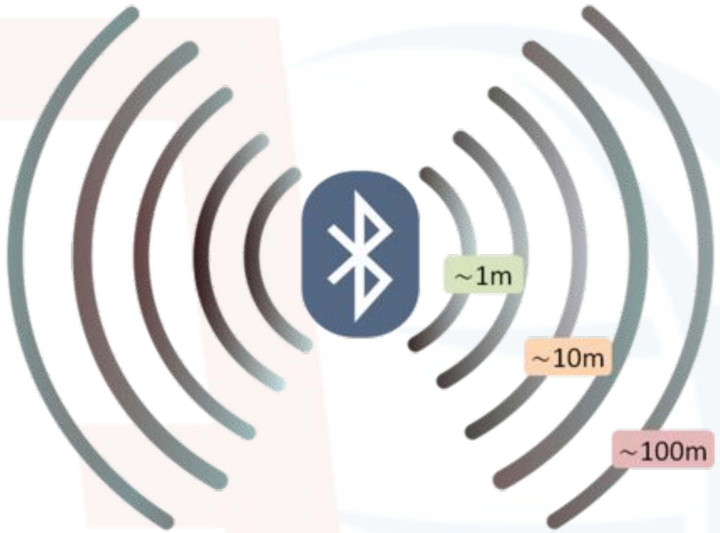
IPS Localization based on Wireless Signal strength

Received Signal Strength Indicator (RSSI)

- **RSSI** is the signal power (or strength) received from a Radio Antenna
  - usually considered for WiFi and BLE, not UWB
- It can estimate the distance *d* of a receiver from an antenna
  - estimation accuracy drops as distance increases
- Heavily affected by interferences



$$d \approx 10^{\frac{\text{measuredPower} - \text{RSSI}}{10N}}$$

measuredPower = RSSI at 1 meter, N = environmental factor

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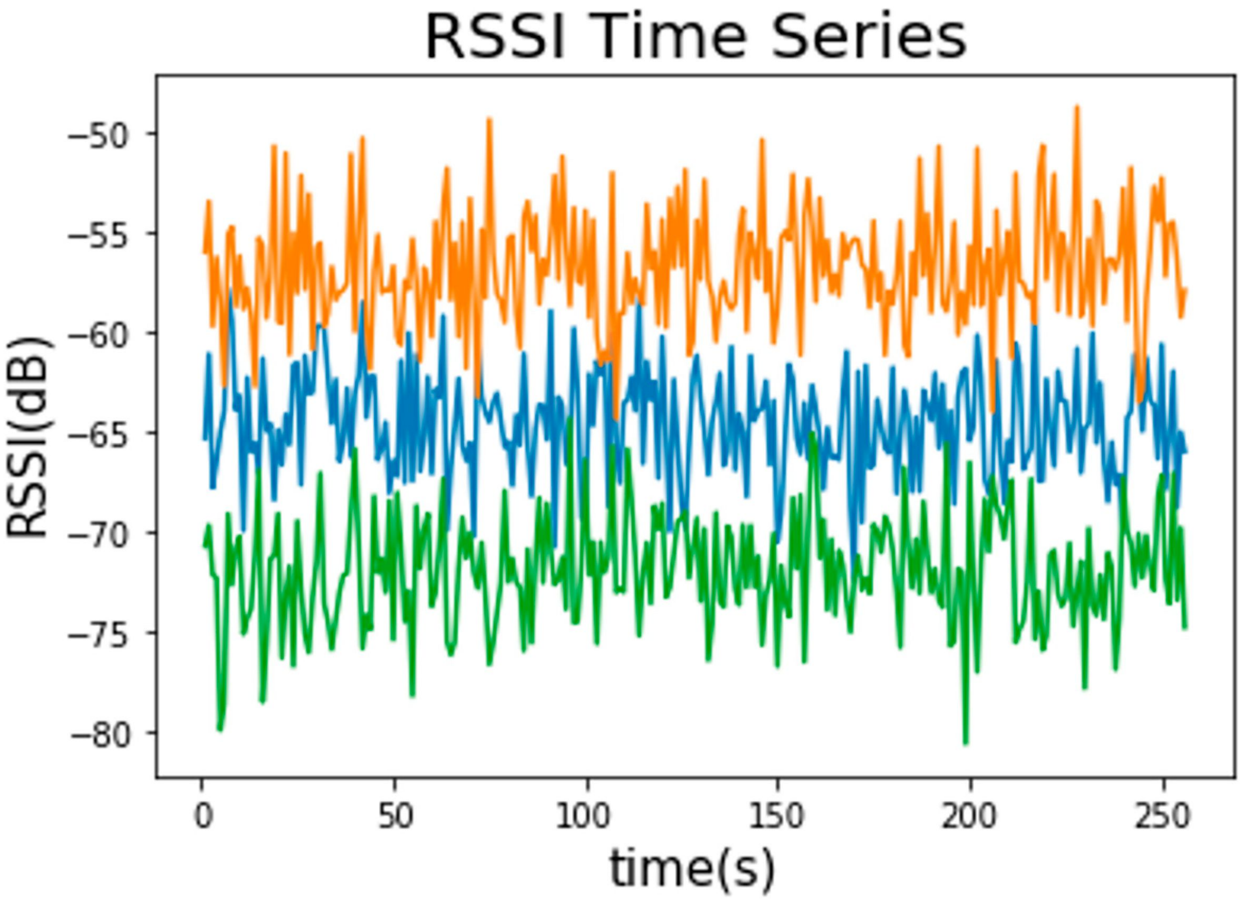
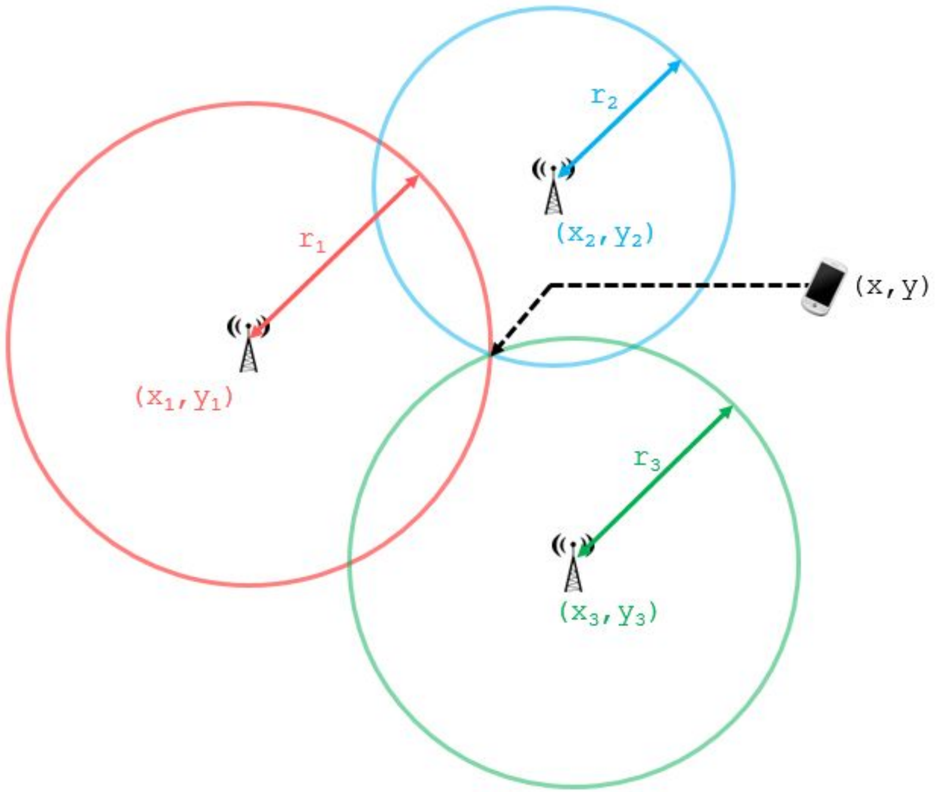
RSSI time series

- RSSI data is expressed in decibels (dB)
  - negative values
    - they represent very small positive energy values in milliWatt, but converted in a logarithmic scale
  - the closer to 0 and the strongest the signal!



Trilateration

- Given three RSSI values, it is possible to use *trilateration* to estimate the position
  - the distance of each RSSI value defines a radium
  - by computing the intersection it is possible to locate the subject
- When more than tree RSSI values are available, it is called **multilateration**



[https://www.101computing.net/cell-phone-trilateration-algorithm]

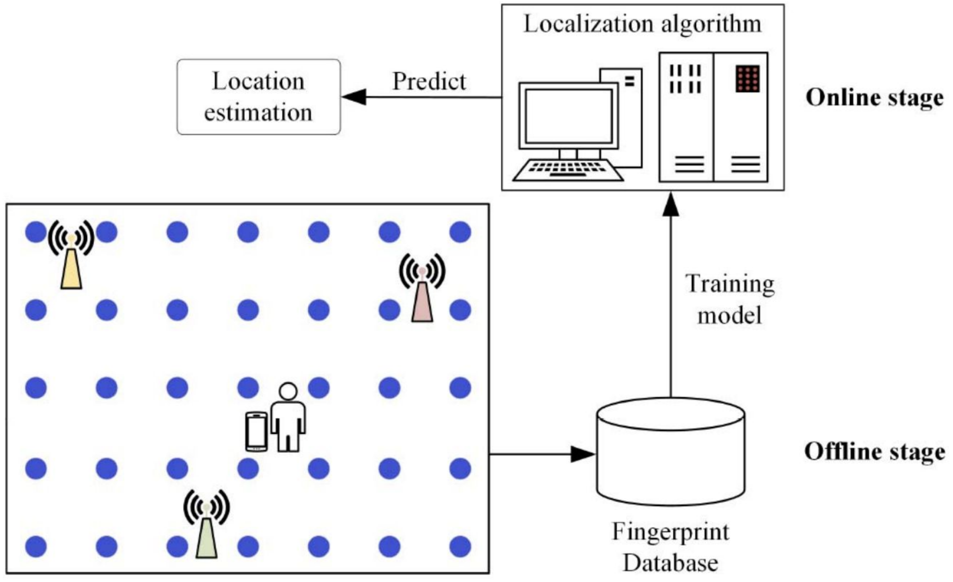
Problems of RSSI distance-based approaches

- Computing distance based on RSSI is not accurate
  - independently from the technology being adopted!
  - this is due to the interferences of the signal in the indoor environment

- Hence, more sophisticated approaches have been proposed

Fingerprinting

- The goal is to collect representative RSSI samples from many positions in the indoor environment.
- The collected samples are annotated with exact coordinates
- Annotated RSSI samples can be used to create a ***radio map***
    - that is then used to estimate the position during inference!



Fingerprinting: the rationale

- For each specific point in the environment, RSSI signals are characteristic
  - ***intuition:*** it is unlikely that the same RSSI values from different antennas are identical in different locations
- Given a position, a ***fingerprint*** is a set of tuples <ID, RSS_{ID}>
  - ID is the antenna identifier
  - RSS_{ID} is the value obtained by antenna ID in that position
    - to increase accuracy, it is also possible to obtain more than one value in each position and perform some aggregation (e.g., avg)

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Fingerprinting: the *OFFLINE* phase

- Fingerprinting requires a training phase, called offline phase
- Someone (a person or a robot) is in charge of moving in the building to:
  - report the current position to the system
  - collect a fingerprint in that position
- This operation is repeated in the whole environment (e.g., at each meter).
  - the result is stored server-side and it is called **radio map**
- A radio map is a set of pairs: ***<Position, Fingerprint>***

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Fingerprinting: the *ONLINE* phase

- During the online phase, the system’s user is in the environment and it requires to compute their position
  - it is unknown and it should be inferred from RSSI data
- Basic approach:
  - choose the fingerprint in the radio map that is the closest to RSSI values collected in current position
  - the corresponding position is chosen as inferred position

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“Naive” Fingerprinting Approach

Minimizing euclidean distance.

$$\hat{x} = \arg\min_{x_j} \left(\sum_{i=1}^{n} (r_i - \rho_i(x_j))^2 \right)$$

inferred position

detected RSSI signal

fingerprint

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Fingerprinting based on Machine Learning

- In practice, in the same position, the fingerprint may change
  - e.g., due to noise, environmental changes, presence of subjects in the environment
- Basic approaches may fail in real-world noisy scenarios
- Machine Learning fingerprinting models are often preferred due to their robustness
  - often it is modeled as a *regression* problem: the goal is to estimate the position!

Examples of models used for Fingerprinting

- K-Nearest Neighbor (the most common “baseline”)
  - matches RSSI signals with K closest fingerprints
  - average the estimated position
- Classic ML algorithms (SVM, Random Forest, Bayesian approaches)
- Deep Neural Networks

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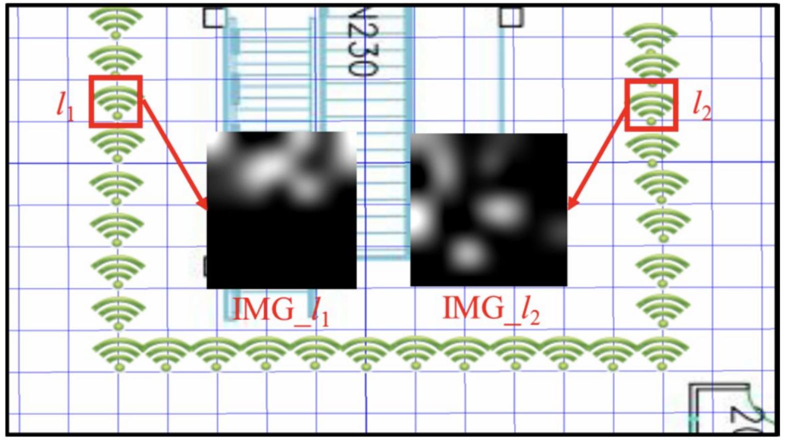
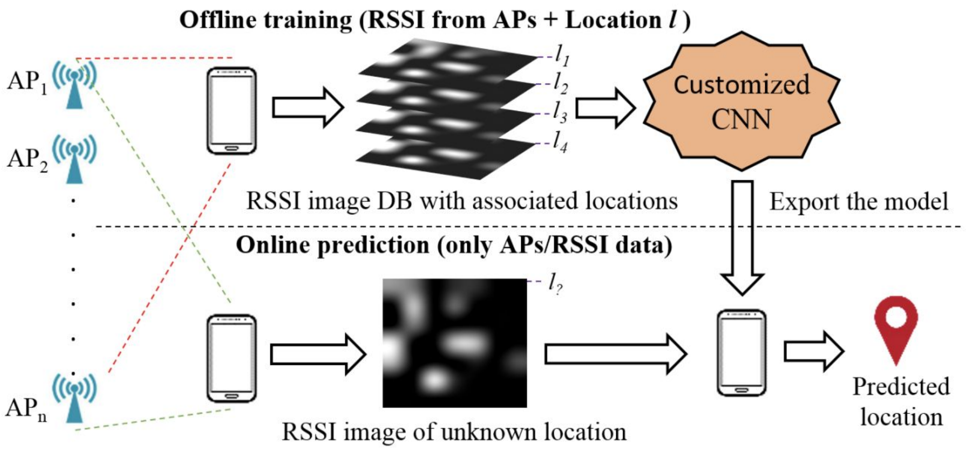
An approach based on CNN

- The RSSI values collected by a device at a specific time instant may be converted into an image!
  - the image represents the map of the indoor environment
  - the intensity of a pixel determines the signal strength for the antennas in a specific location (normalized RSSI value)





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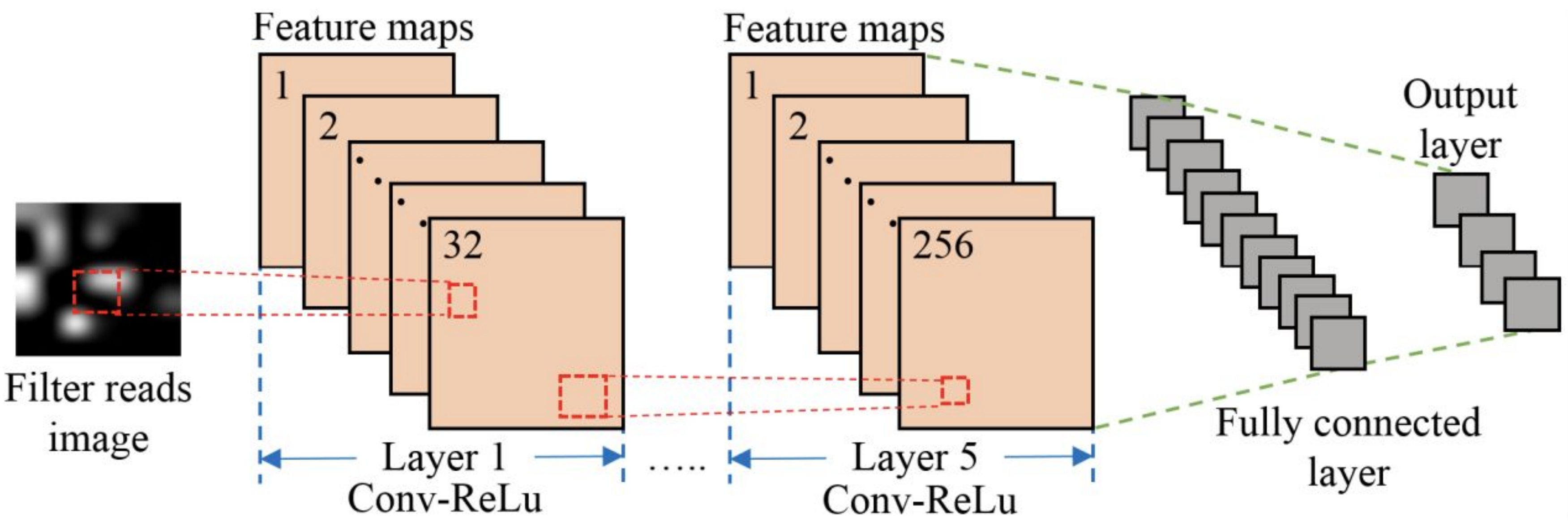
Mittal, A., Tiku, S., & Pasricha, S. (2018, May). Adapting convolutional neural networks for indoor localization with smart mobile devices. In *Proceedings of the 2018 on Great Lakes Symposium on VLSI* (pp. 117-122).

Using CNNs for Indoor Localization (cont.)



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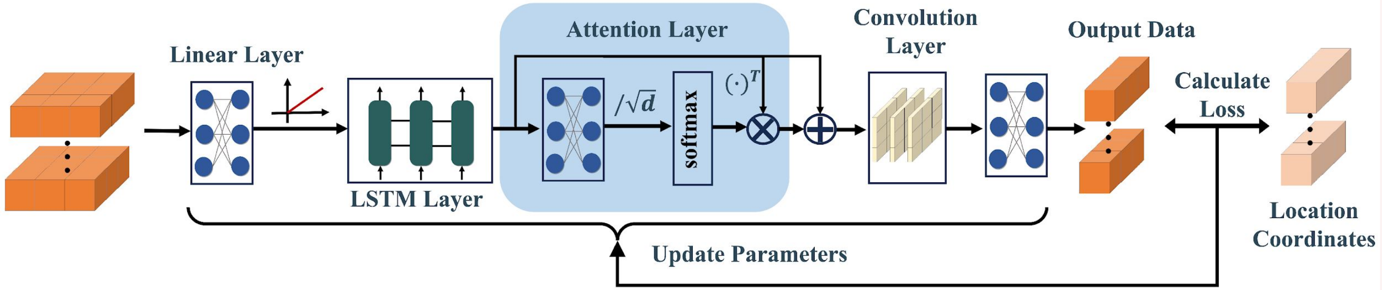
An approach based on LSTMs, self-attention, and CNN

- Each input is a ***sequence*** of fingerprints
- LSTM + Self-Attention mechanism to map sequences into embeddings
- This method also leverages a CNN layer to capture spatial properties from LSTM embeddings
  - we will see also in other lessons that combining LSTMs and CNNs for time series is very common and effective!



[Wu, Z., Hu, P., Liu, S., & Pang, T. (2024). Attention mechanism and LSTM network for fingerprint-based indoor location system. Sensors, 24(5), 1398.]

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Channel State Information (CSI)

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Channel State Information (CSI)

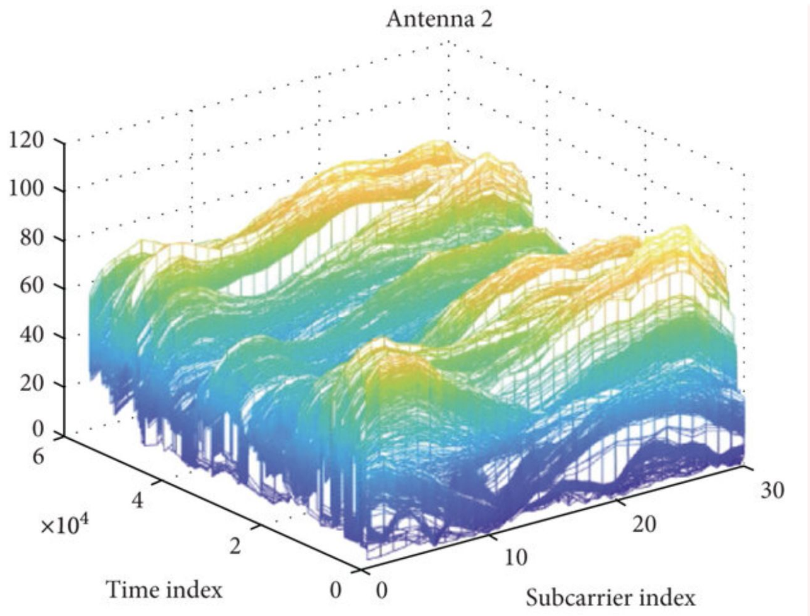
- Alternative to RSSI
  - While RSSI is widely used due to its simplicity, it is susceptible to multipath effects and interference
  - CSI is obtained from WiFi access points and mitigates these problems

- CSI encodes the characteristics (phase and amplitude) of the received signal at different frequencies
  - implicitly capturing the scattering and refraction encountered by the signal during transmission

- While it is more robust than RSSI, only a few off-the-shelf receiver devices can actually compute it

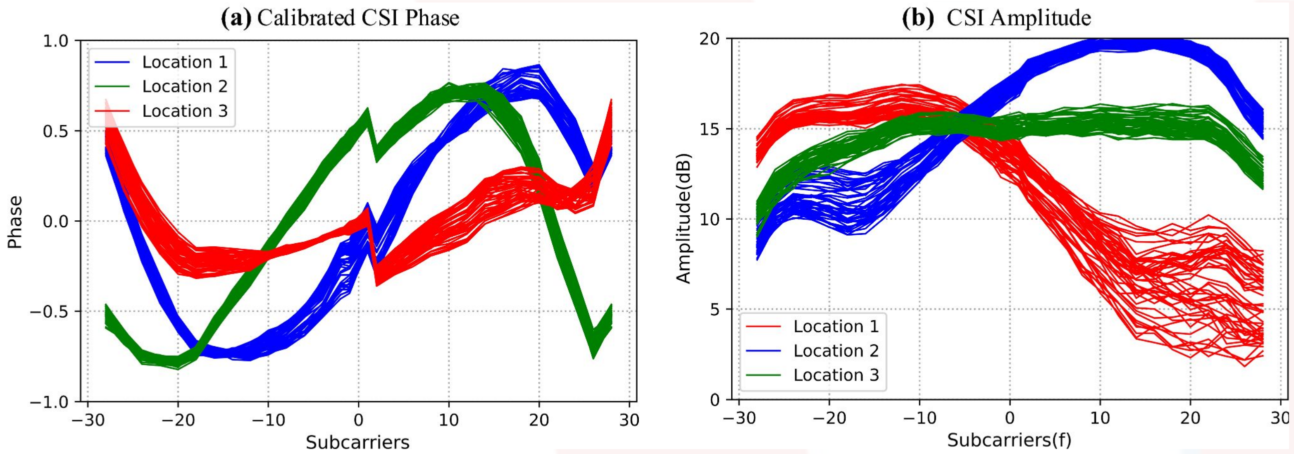
Orthogonal Frequency-Division Multiplexing (OFDM)

- The transmitter (usually WiFi APs) encodes, in a single channel, each message on multiple subcarrier frequencies
- Subcarriers are used to transmit data over multiple, closely spaced frequencies, reducing interference and optimizing the use of the available bandwidth
- For each subcarrier, we can also obtain information about how the signal propagates
  - i.e., amplitude and phase
- The receiving device (i.e., the device to be localized) requires an OFDM demodulator to decompose the received signal into its constituent subcarriers



CSI for Indoor Localization

- Each subcarrier is an oscillating wave at a specific frequency
  - The multipath effect may impact in a different way each frequency!
- The transmitter broadcasts a “standard” message
- The receiver demodulates the message and derives phase and amplitude for each subcarrier
  - the *phase* intuitively is the 'arrival point' of the subcarrier signal at the receiver, indicating where the wave is in its cycle when it arrives
  - the *amplitude* intuitively reflects the intensity of the signal when it reaches the receiver
- Each location in an indoor environment has its unique patterns!
  - indeed, distance and interferences affect the received amplitude and phase of each frequency in a specific way



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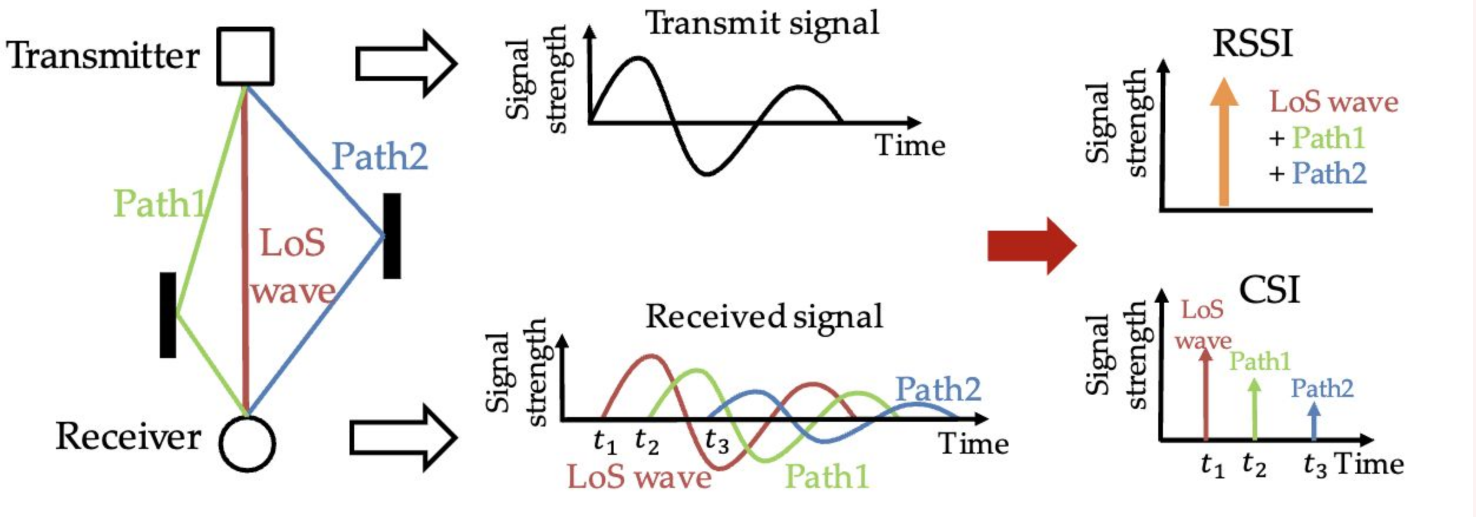
RSSI vs CSI

Since each subcarrier (with a specific frequency) can be affected in different ways by the multipath effect, we can have more reliable information than only using RSSI!



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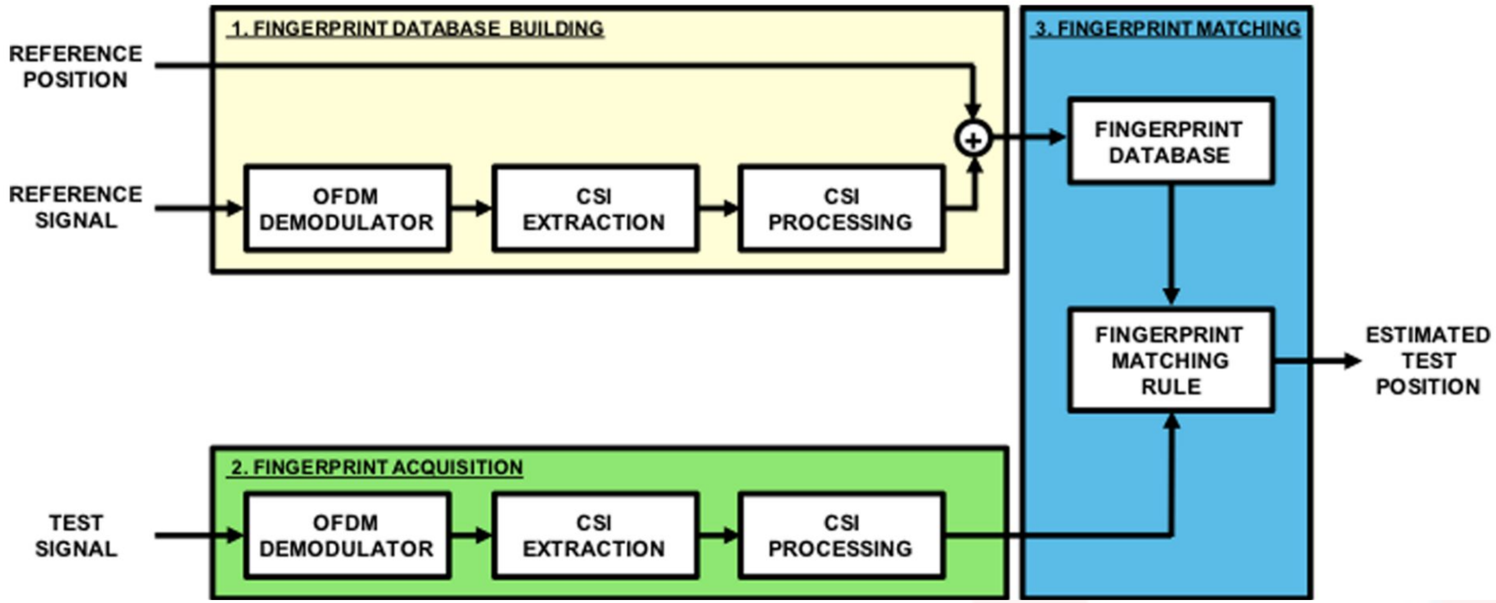
CSI Fingerprinting

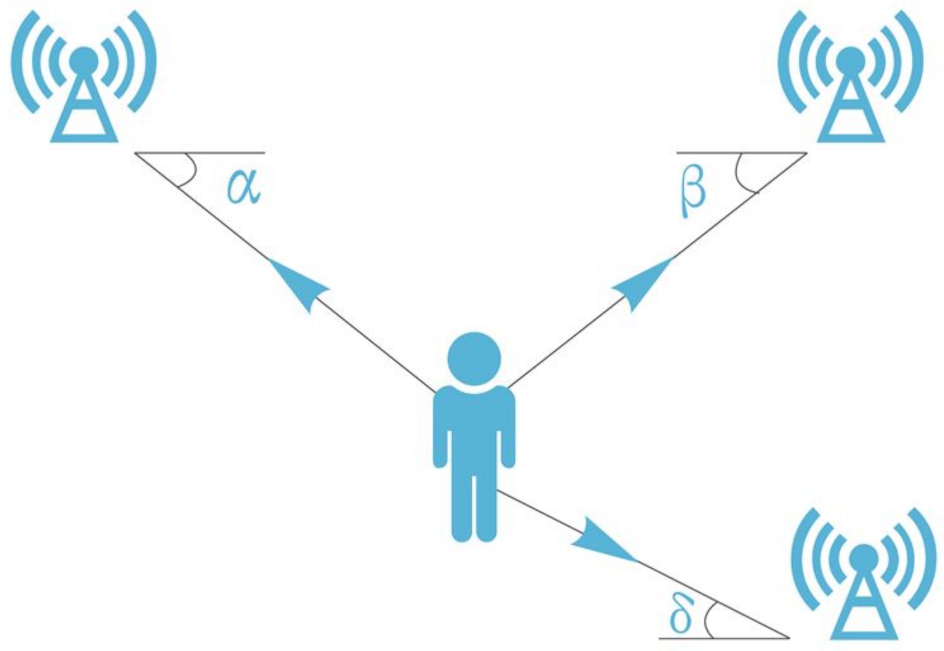
- CSI-based localization is usually done with supervised fingerprinting
- Hence, similar techniques to the one that we have seen for RSSI signals!



Other IPS Localization approaches

Angle of Arrival (AoA)

- Advanced anchors (e.g., UWB) in the environment can also compute the ***angle*** with respect to the receiver device
  - through directionally sensitive antennas
- Trilateration (or multilateration) achieves sub-meters positioning accuracy!
  - combining distances and angles values
- Drawback: AoA requires sophisticated and costly hardware



https://www.minew.com/angle-of-arrival-minew-stretches-out-its-antennae-to-higher-precision-positioning/

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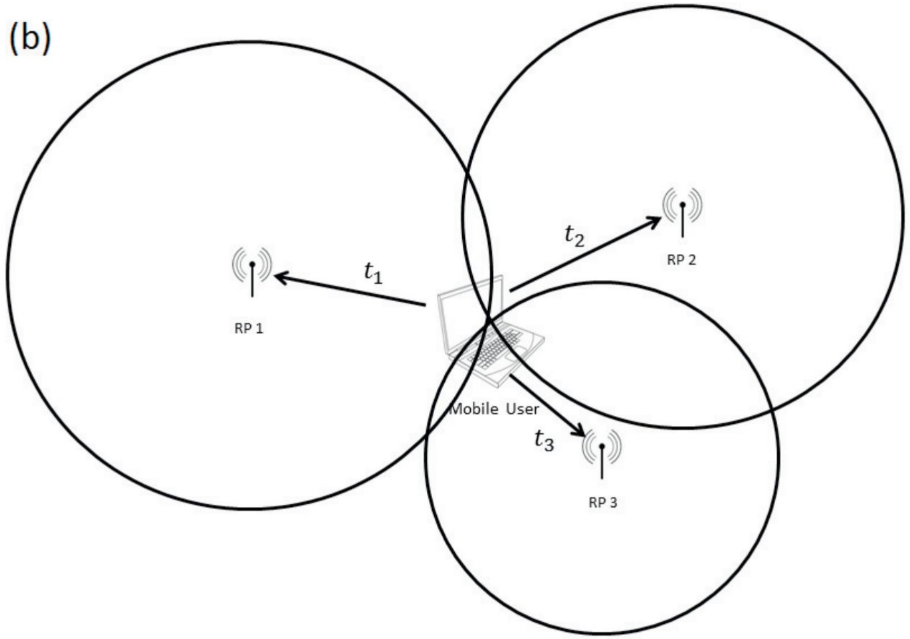
Time of Arrival (ToA) / Time of Flight (ToF)

- Measuring the ***absolute travel time*** of signal from transmitter to receivers to estimate distance!
  - *travelTime * waveSpeed*
  - Wave speed heavily depends on the material that signal has to penetrate
  - it requires knowing the position of each antenna
- Requires precise synchronization between clocks (between antennas and receiver!)
- Works well when receiver and transmitters are in Line of Sight and without obstacles
- Usually implemented with UWB



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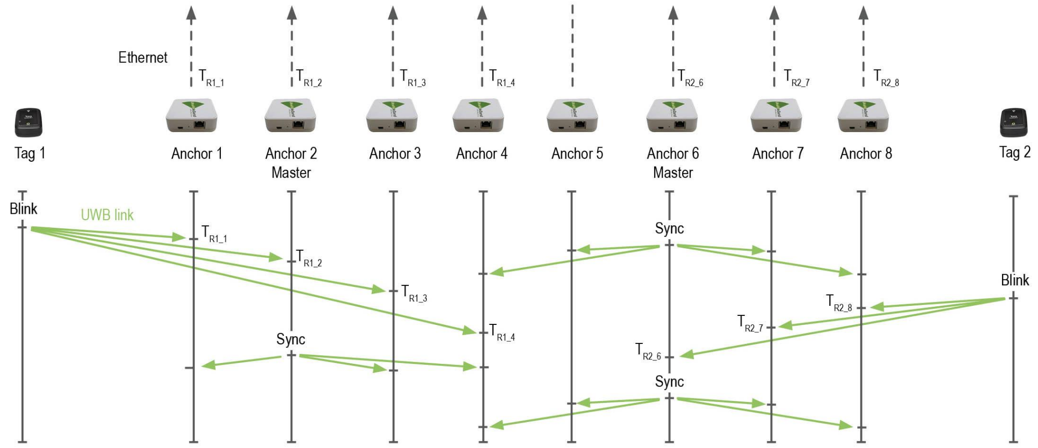
Time Difference of Arrival (TDoA)

- Improved version of ToA trying to remove bias from the clock of the mobile device
- For each pair of antennas, the time difference at which a message is received is computed
- Time differences are combined to localize the mobile device
  - since we know the exact positions of each antenna, we can apply multilateration
- ***Drawback***: antennas should be perfectly synchronized between them!



Summary: Pros and Cons

Technique	Advantages	Disadvantages
RSSI	Easy to implement, cost efficient, can be used with a number of technologies	Prone to multipath fading and environmental noise, lower localization accuracy, can require fingerprinting
CSI	More robust to multipath and indoor noise,	It is not easily available on off-the-shelf NICs
AoA	Can provide high localization accuracy, does not require any fingerprinting	Might require directional antennas and complex hardware, requires comparatively complex algorithms and performance deteriorates with increase in distance between the transmitter and receiver
ToF	Provides high localization accuracy, does not require any fingerprinting	Requires time synchronization between the transmitters and receivers, might require time stamps and multiple antennas at the transmitter and receiver. Line of Sight is mandatory for accurate performance.
TDoA	Does not require any fingerprinting, does not require clock synchronization among the device and RN	Requires clock synchronization among the RNs, might require time stamps, requires larger bandwidth
Fingerprinting	Fairly easy to use	New fingerprints are required even when there is a minor variation in the space

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Inertial Based Approaches

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Dead Reckoning

- Inertial sensors in mobile devices can capture the acceleration in a specific direction
  - combining accelerometer and gyroscope
- Starting from a position obtained through indoor localization approaches (e.g., radio-based, computer vision) it is possible to estimate displacements!
  - useful when indoor positioning information is not constantly available, to estimate short displacements
- For instance, ***step detection*** is often leveraged to estimate displacements

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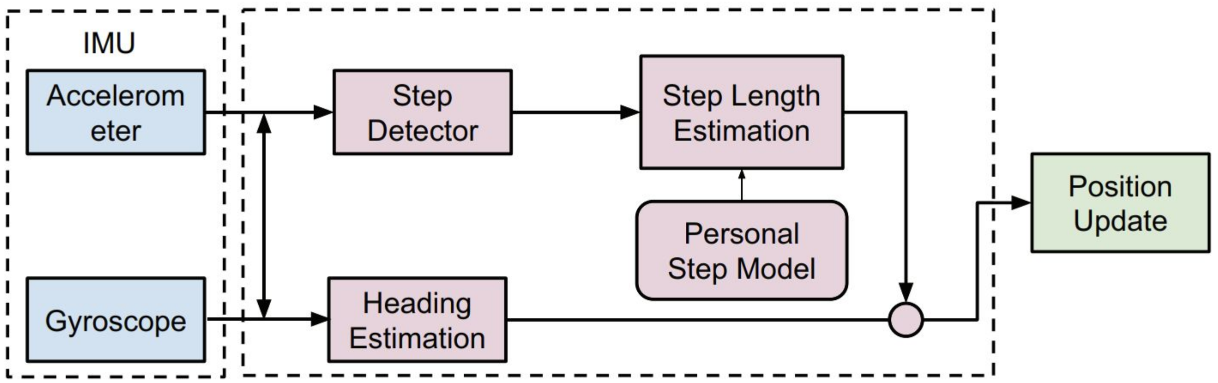
Dead Reckoning pipeline

- Using inertial sensors to estimate steps, estimate stride length and heading.
- Goal: updating the position obtained by external reference
  - e.g., RSSI
- A personalized step model may help in further improving the approximation
- The position update can be done analytically or by using deep learning models!



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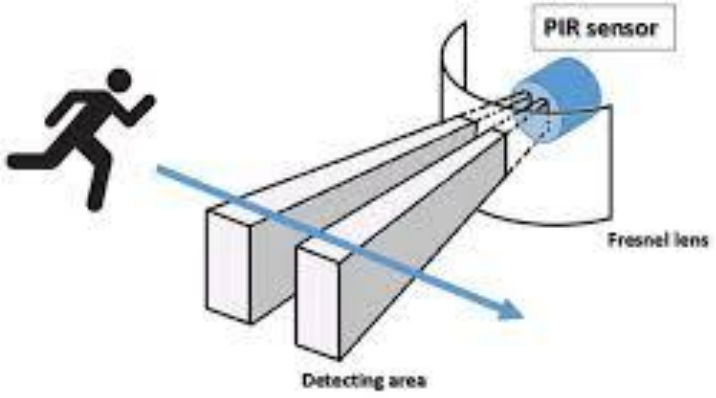
Problems of Dead Reckoning

- Since inertial sensors are often noisy is possible to wrongly estimate the displacement
    - error in counting steps and their length
    - errors in computing the direction
- The error in computing new positions (*drift)* grows linearly with time
    - even small approximation errors at the begin have significant impact as time passes by
    - hence dead reckoning should be used to compute small displacements
    - that’s why dead reckoning can not be used alone without any other localization technique

Alternative IPS technologies

Passive InfraRed (PIR)

- PIR sensors can detect human motions/proximity in specific areas
- Users not requiring wearing additional devices
- Many drawbacks:
  - coarse-grained localization
  - a sensor for each area of interest
  - challenging with more than one users
    - while it is possible to estimate the number of subjects in an indoor environment with PIR sensors, it is not possible to track personalized locations



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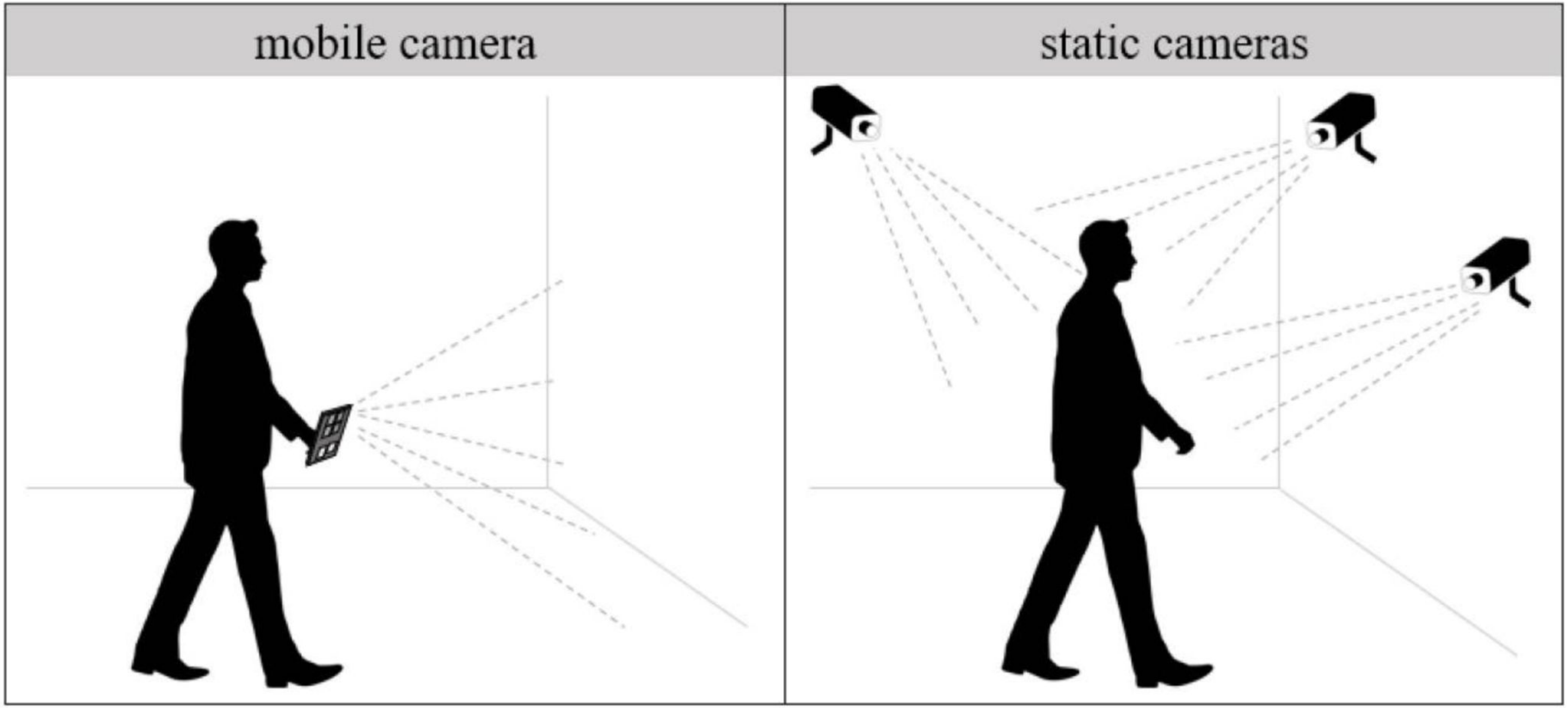
Computer Vision



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Static VS Mobile Cameras

- **Mobile Camera**
  - markers (explicit or implicit) in the environment are needed to compute position
  - it requires the user to actively use the device to frame the markers
    - but it can be complemented with dead reckoning!
  - a good solution for ***navigation*** in smart buildings
- **Static Camera**
  - passive approach, user does not require to use device
  - easier to implement
  - may be perceived as too intrusive in terms of privacy
  - a good solution for ***surveillance***

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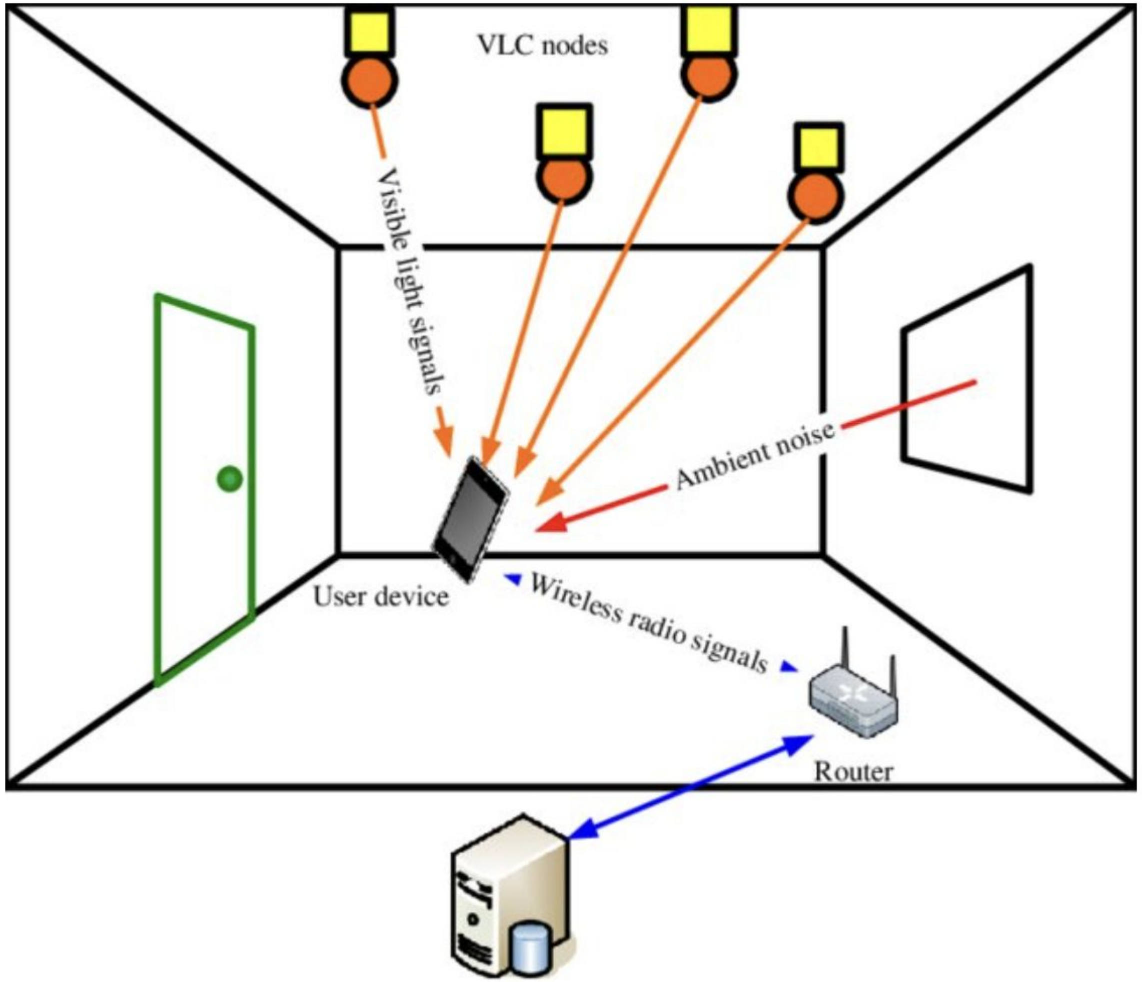
Visible Light Communication (VLC)

- Emerging technology for high-speed data transfer
- Data emitted by LEDs
- The receiver uses a light sensor to estimate position and direction of LED emitters
- **Pro**: wide-scale proliferation of this technology
- **Cons:** it requires Line of Sight (LoS)



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